The centre fitted on the tailstock spindle is called dead centre. It is useful in supporting the other end of the work. Centres are made of high carbon steel and hardened and then tempered. So the tips of the centres are wear resistant. Different types of centres are available according to the shape of the work and the operation to be performed.

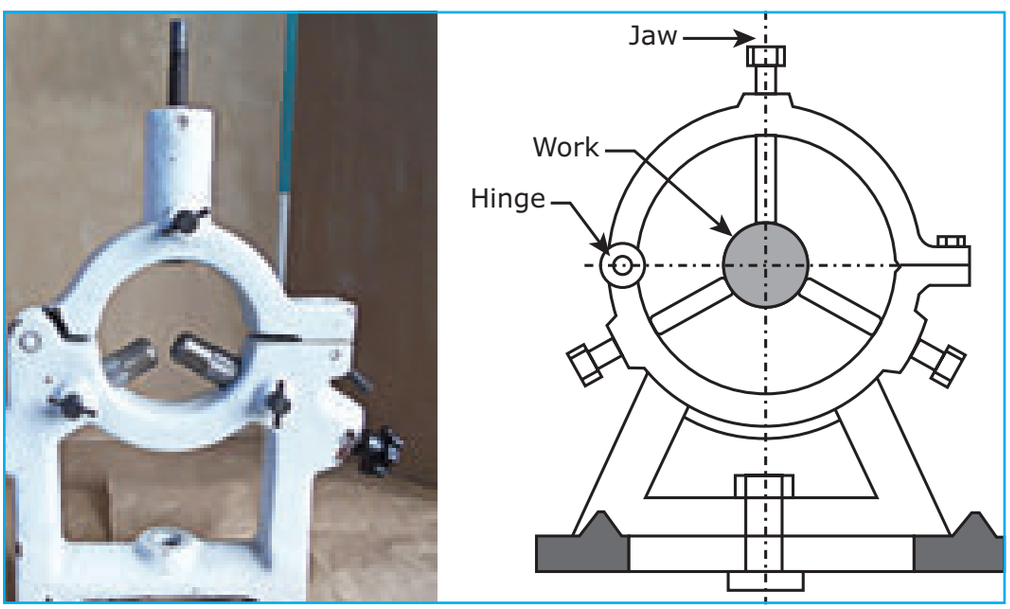
Rests

A rest is a mechanical device to support a long slender workpiece when it is turned between centres or by a chuck. It is placed at some intermediate point to prevent the workpiece from bending due to its own weight and vibrations setup due to the cutting force. There are two types of rests

1. Steady rest
2. Follower rest

Steady rest

Steady rest is made of cast iron. It may be made to slide on the lathe bedways and clamped at any desired position where the workpiece need support. It has three jaws. These jaws can be adjusted according to the diameter of the work. Machining is done upon the distance starting from the headstock to the point of support of the rest. One or more steady rests may be used to support the free and of a long work.



Steady Rest

Follower rest

It consists of a 'C' like casting having two adjustable jaws to support the workpiece. The rest is bolted to the back end of the carriage. During machining, it supports the work and moves with the carriage. So, it follows the tool to give continuous support to the work to be able to machine along the entire length of the work.

Note: In order to reduce friction between the work and the jaws, proper lubricant should be used.

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